Security Operations Center

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Problem Statement

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“Why Do We need SOC?

What issue will SOC solve for the organization?

What must the SOC accomplish to solve the existing problems?

Executive and Board Support

Beyond
Imagination

36%

of respondents say that their SOC does not interact with the business.

22% of respondents do not know if their SOC interacts with the business.

20%

of respondents say that the SOC receives **annual** updates from the business to understand and address their concerns and risks.

10% recieve **quarterly** updates.

Introduction

A security operations center, or SOC, is a team of expert individuals and the facility in which they dedicate themselves entirely to high-quality IT security operations.

A security operations center (**SOC**) is a facility that houses an information security team responsible for monitoring and analyzing an organization's security posture on an ongoing basis.

Beyond Imagination

Introduction

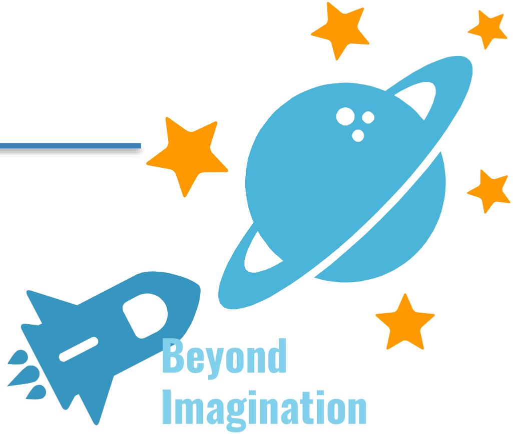
Beyond Imagination

The SOC team's goal is to detect, analyse, and respond to cyber security incidents using a combination of technology solutions and a strong set of processes.

Security operations centers are typically staffed with security analysts and engineers as well as managers who oversee security operations.

SOC staff work close with organizational **incident response** teams to ensure security issues are addressed quickly upon discovery.

Introduction



Security operations centers monitor and analyze activity on networks, servers, endpoints, databases, applications, websites, and other systems, looking for anomalous activity that could be indicative of a security incident or compromise.

The SOC is responsible for ensuring that potential security incidents are correctly identified, analyzed, defended, investigated, and reported.

Setting Up the SOC

THERE ARE TWO CRITICAL FUNCTIONS IN BUILDING A SOC

Beyond Imagination

1

The first is setting up your security monitoring tools to receive raw security-relevant data (e.g. login/logoff events, persistent outbound data transfers, firewall allows/denies, etc.).

This includes making sure your critical cloud and on-premises infrastructure (firewall, database server, file server, domain controller, DNS, email, web, active directory, etc.) are all sending their logs to your log management, log analytics, or SIEM tool.

Setting Up the SOC

THERE ARE TWO CRITICAL FUNCTIONS IN BUILDING A SOC

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2

The second function is to use these tools to find suspicious or malicious activity by analyzing alerts; investigating indicators of compromise (IOCs like file hashes, IP addresses, domains, etc.); reviewing and editing event correlation rules; performing triage on these alerts by determining their criticality and scope of impact; evaluating attribution and adversary details; sharing your findings with the threat intelligence community; etc.

Difference Between CSIRT and SOC Team

CSIRT also Called as CERT or CIRT

The core function of a CSIRT is to minimize and manage damage caused by an incident, the CSIRT does not just deal with the attack itself, they also communicate with clients, executives, and the board.

Roles of SOC Team

A SOC team comprises several roles:

1. Security analyst : - responsible for detecting potential security threats and handling them. Also implements security measures and is involved in disaster recovery plans.

2. Security engineer: - in charge of maintaining and updating tools and systems and is usually a software or hardware specialist. They are also responsible for any documentation that might be needed by other team members, such as protocols.

Roles of SOC Team

A SOC team comprises several roles:

3. SOC manager: - directs SOC operations, responsible for the SOC team. Responsible for syncing between analysts and engineers, hiring, training, and security strategy. Directs and orchestrates response to major security threats.

4. Chief information security officer (CISO): - establishes security related strategies, policies, and operations. Works closely with the CEO, informs and reports to management on security issues.

Roles of SOC Team

A SOC team comprises several roles:

5. Director of incident response : - responsible for managing incidents in large companies as they occur and communicating security requirements to the organization in the case of a significant breach.

How do S-O-C work?

An organization must first define its security strategy and then provide a suitable infrastructure for the SOC team to work with.

The information system that underlies SOC activity is a security information and event management (SIEM) system, which collects logs and events from hundreds of security tools and organizational systems, and generates actionable security alerts, which the SOC team can analyze and respond to.

SOCs use strategic methodologies and processes to build and maintain the company’s cybersecurity defenses

How do S-O-C work?

Task of Security Operation Center’s Team:

1. Establishing awareness of assets – From the start of their operations, SOCs need to be well-versed in the tools and technologies at their disposal, as well as the hardware and software running on the network.

2. Proactive monitoring – Instead of focusing on reactive measures if irregularities occur, SOCs take intentional steps to detect malicious activities before they lead to substantial harm.

How do S-O-C work?

Task of Security Operation Center’s Team:

3. Managing logs and responses – In case of a breach, it is essential to be able to retrace your steps to find where something may have gone wrong.

4. Ranking alerts – When irregularities surface, one of the tasks a SOC will undergo is ranking the severity of incidents.

5. Adjusting defenses – Vulnerability management and increasing the awareness of threats are essential parts of preventing security breaches.

How do S-O-C work?

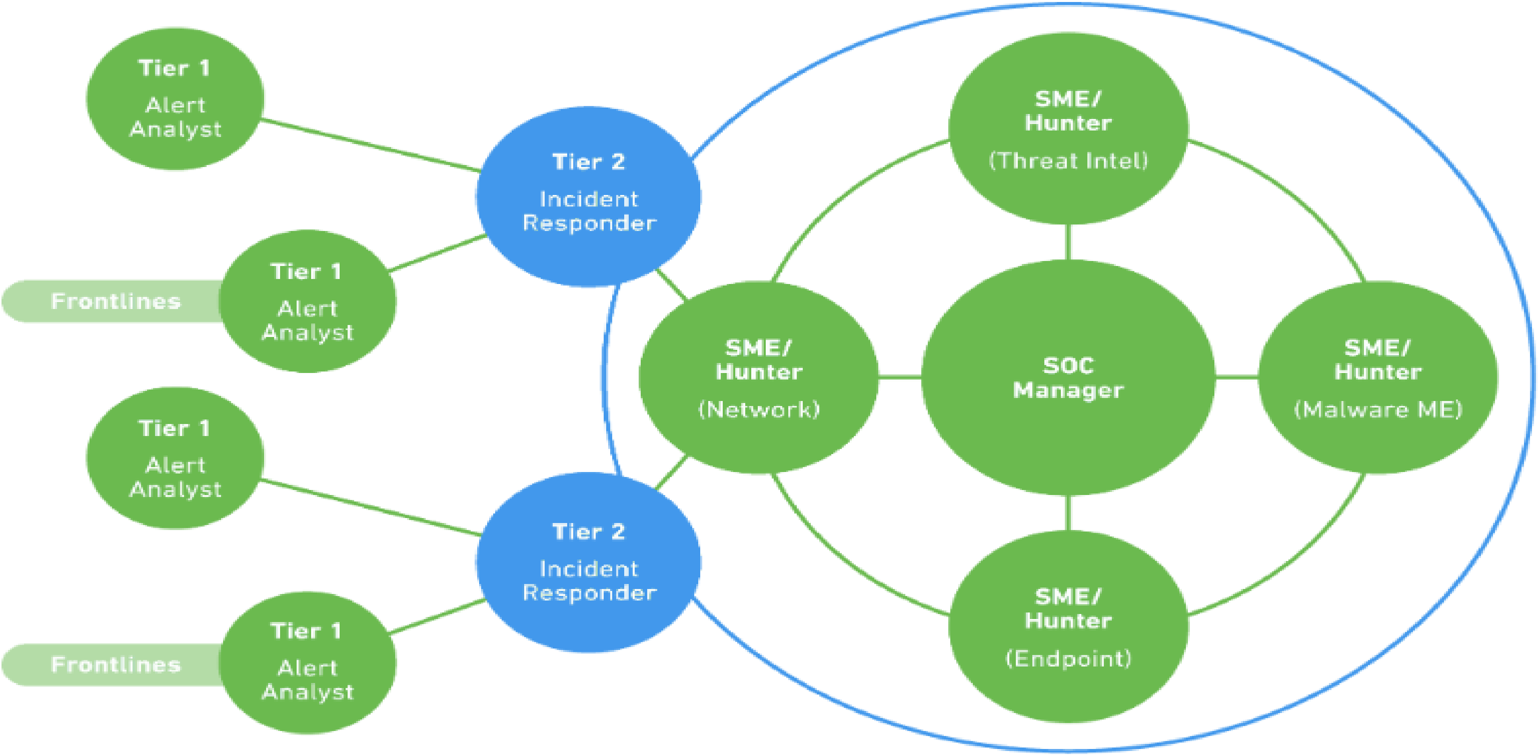
Task of Security Operation Center’s Team:

6. Checking compliance – In this age of data technology, there are few things more relevant to information security than maintaining essential compliance regulations.

SOCs use their daily efforts to keep up with any mandatory protective measures while going a step further to keep the company from harm.

Roles and Tiers of SOC

SOC analysts are organized in four tiers:



Roles and Tiers of SOC

SOC analysts are organized in four tiers:

1. SIEM alerts flow to Tier 1 analysts who monitor, prioritize and investigate them.

2. Real threats are passed to a Tier 2 analyst, with deeper security experience, who conducts further analysis and decides on a strategy for containment.

Roles and Tiers of SOC

SOC analysts are organized in four tiers:

3. Critical breaches are moved up to a Tier 3 senior analyst, who manages the incident. Tier 3 analysts are also responsible for actively hunting for threats and assessing the vulnerability of the business.

4. The Tier 4 analyst is the SOC manager, in charge of recruitment, strategy, priorities and the direct management of SOC staff when major security incidents occur.

Benefits of SOC

Incident response—SOCs operate around the clock to detect and respond to incidents.

Threat intelligence and rapid analysis—SOCs use threat intelligence feeds and security tools to quickly identify threats, and fully understand incidents to enable appropriate response.

Reduce cybersecurity costs—although a SOC represents a major expense, in the long run it saves the costs of ad hoc security measures and the damage caused by security breaches.

Reduce the complexity of investigations—SOC teams can streamline their investigative efforts.

Challenges Faced by SOC

63%

of GISS respondents cite budget constraints as the main challenge to the Information Security function's contribution to the organization.

53%

also cite the lack of skilled resources as a barrier to value creation.

Challenges Faced by SOC

Increased volumes of security alerts—the growing number of security alerts requires a significant amount of an analyst's time.

Management of many security tools—as various security suites are being used by SOCs and CSIRs, it is hard to efficiently monitor all the data generated from data points and sources.

Resource allocation—staffing or lack of qualified individuals is an issue. An organization may decide to outsource, however, the issue of greater vulnerability that comes with remote working conditions arises.

“THANKS!”